AIORI HACKATHON 3 / PROJECT PS-011

Authorised Attack-Surface Inventory & Change Monitor

A practical workflow for discovering, tracking and reviewing changes across an approved asset inventory.

TEAM	TEAM CODE	SUBMITTER	INSTITUTE
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Project at a glance

A bounded inventory, four documented checks, baseline comparison, analyst review, and exportable evidence.

Verification note: 18 unit tests and 4 demo tests passed in the reported Windows session. Seeded findings are synthetic test scenarios, not live vulnerability claims.

01 | Why this problem matters

Digital assets change over time. Without a repeatable inventory, small changes may be missed.

ASSET DRIFT

Domains, DNS records and exposed services can change between reviews.

TLS EXPIRY

Expired certificates can affect trust, browser warnings and service availability.

UNEXPECTED SERVICES

A reachable port may need an owner to verify whether it is intended.

DNS UNCERTAINTY

A DNS record that points to a missing provider resource is a candidate for investigation—not proof of exploitability.

The project goal

Turn scattered observations into a repeatable, scoped inventory with clear evidence and a human review path.

02 | Objectives & authorised scope

Designed for controlled monitoring—not unrestricted scanning.

20	10	4	0
maximum lab domains	maximum lab hosts	documented checks	exploitation steps in demo

IN SCOPE

Written, organiser-approved domains and hosts; DNS observations; TLS certificate status; selected TCP ports; baseline comparisons; findings review; reports and CSV exports.

OUT OF SCOPE

Unapproved targets, unrestricted scanning, exploitation, credential attacks, and declaring potential dangling DNS exploitable without contextual validation.

03 | How the system works

A simple flow from approved assets to reviewed, explainable findings.

01	02	03	04	05	06
Approved scope	**Scope validation**	**Observation checks**	**Data storage**	**Baseline compare**	**Review & export**
Domains + hosts	Reject out-of-scope targets	DNS • TLS • selected ports	Timestamped observations	Detect changes and drift	Dashboard • findings • CSV

Technology overview

Python application • Flask dashboard • SQLite-backed observations • DNS/TLS/port check modules • baseline/change detector • report and CSV exports.

Use the live application during the presentation to demonstrate the actual execution path; this slide is a high-level architecture view.

04 | Four documented checks

Each check contributes evidence to the inventory and change-review process.

01 DNS MONITORING

Purpose: record configured DNS observations and lookup outcomes.

Output: DNS records and potential dangling-DNS candidates for contextual review.

02 TLS CERTIFICATE CHECK

Purpose: inspect certificate validity and expiry for configured targets.

Output: certificate status and expiry findings.

03 SELECTED TCP PORTS

Purpose: check only configured ports on approved hosts.

Output: reachability/service observations for the selected ports.

04 BASELINE CHANGE DETECTION

Purpose: compare current observations with a stored baseline.

Output: recorded changes and findings that can be reviewed.

05 | Findings & analyst triage

The prototype uses seeded scenarios to demonstrate detection and review.

SEEDED SCENARIO	EXPECTED DEMONSTRATION	RESPONSIBLE INTERPRETATION
Expired TLS certificate	Expiry finding with supporting scan evidence	Controlled test case; confirm it against the scan record.
Unexpected service	Unexpected port/service finding	An owner must confirm whether the service is authorised.
Potential dangling DNS	Candidate finding with status and review notes	A candidate alone is not proof of exploitability.

Why triage matters

Analyst statuses and notes help separate new observations from acknowledged, resolved, or false-positive cases. Re-observed findings should preserve review context rather than multiplying duplicate active alerts.

06 | Live prototype walkthrough

Suggested 4–5 minute demo sequence for judges.

STEP	PAGE / ACTION	WHAT TO EXPLAIN
1	Overview	Show the bounded inventory and current/latest findings.
2	Seeded demo / Scan History	Explain that seeded scenarios are synthetic test evidence, not live external vulnerability claims.
3	DNS, TLS and Port pages	Show one example from each check and point to recorded evidence.
4	Security Findings	Explain severity, confidence, notes, and false-positive review.
5	Change History	Explain how comparing with a baseline highlights drift.
6	Reports & Exports	Generate/open a report and show the available CSV exports.

Tip: keep the browser at a readable zoom, close unrelated windows, and rehearse the transitions before presenting.

07 | Verification evidence

The following results were reported from the project's Windows PowerShell session.

18	4	c87b646
unit tests passed	demo tests passed	latest pushed commit

Commands and outcomes

`python -m unittest discover tests` → 18 tests, OK
`python main.py test` → 4 tests, OK
`git push origin master` → commit c87b646 pushed to origin/master.

Automated tests validate selected scenarios; they do not replace evidence from authorised checks or a rehearsed end-to-end demo.

08 | Safety, limitations & responsible interpretation

A useful security report must clearly separate observations from conclusions.

AUTHORISED SCOPE

Only written, organiser-approved lab targets should be checked.

NON-EXPLOITATION

The project focuses on inventory and diagnostic observations, not exploitation.

SYNTHETIC DATA

Label seeded findings as simulated test scenarios.

HUMAN REVIEW

Unexpected services and potential dangling DNS require contextual validation.

EVIDENCE INTEGRITY

Use genuine screenshots, current reports, timestamps and actual test results.

09 | What remains before submission

Use this checklist to turn the prototype into a clear, evidence-backed submission.

01	**Capture real dashboard screenshots** Overview • findings • scan history • reports
02	**Review final report PDF** Use the required organiser template; remove placeholders.
03	**Rehearse the demo** Start cleanly, run tests, and explain seeded vs live data.
04	**Confirm submission requirements** Repository access • team video • presentation/report files • portal status

Update each item only when you have completed it and can show the supporting evidence.

Thank you

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> **Key takeaway**
>
> A strong monitoring workflow combines an approved scope, repeatable observations, baseline comparison, and an explainable human review process.

Repository: https://github.com/aiori-hackathon/Cyber-Warriors

Presentation working version: add genuine dashboard screenshots and confirm official submission formatting before treating it as final.